Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/28/2014
PI Name:	Nicholson, Wayne Ph.D.		
Project Title:	Development of Multiple Antibiotic Resistance by Opportunistic Bacterial Pathogens during Human Space Flight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	32953	**Congressional District:**	3
Comments:	NOTE: Congressional District is for University of Florida (32611-0001)--Ed., 8/17/2012; changed to 3 (Ed., 7/30/13)		
Project Type:	Flight	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	07/27/2012	**End Date:**	09/30/2015
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date is now 9/30/2015 per NSSC information (Ed., 4/14/15) NOTE: New end date of 4/30/2015 (original end date was 7/26/13), per NSSC information (Ed., 4/14/14) NOTE: new end date of 4/30/2014 (original end date was 7/26/13), per NSSC information (Ed., 8/20/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fajardo-Cavazos, Patricia (University of Florida)		
Grant/Contract No.:	NNX12AN70G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Background: Maintaining astronaut health during long-term spaceflight is of prime importance to the success of exploration missions to destinations such as the Moon, near-Earth asteroids, or Mars. The immune system weakens during missions in Low Earth Orbit (LEO), rendering astronauts more susceptible to infection. Microgravity induces in certain bacterial species increased virulence properties such as: adherence to mammalian cells, biofilm formation, resistance to acid and macrophages, and antibiotic resistance. As part of the normal population of microbial inhabitants on and within the human body, astronauts carry many microbes considered “opportunistic pathogens”, i.e. normally harmless microbes with the potential to cause disease in hosts with lowered immune function. Opportunistic infections are often treated with combinations of two antibiotics that differ in their mechanisms of action. Combination treatment can be effective because (i) certain antibiotic combinations work synergistically to more effectively combat the pathogen than either drug given alone, and (ii) the probability of the pathogen evolving resistance to two antibiotics simultaneously is much lower than that of becoming resistant to a single antibiotic. Rationale: Susceptibility of potential pathogens to single antibiotics in the spaceflight environment has been studied to a limited extent, but a systematic study of the development of multiple antibiotic resistance by mutation has not to date been attempted. In a previous space exposure experiment (PROTECT on ISS using the bacterium Bacillus subtilis), we found that certain mutations to rifampicin (RIF) resistance also resulted in resistance to trimethoprim (TMP). RIF and TMP are prescribed in combination for treatment of certain infections caused by the important opportunistic pathogen Staphylococcus aureus. Goal: We propose to test the hypothesis that exposure to the human spaceflight environment, particularly microgravity, increases the development of simultaneous resistance to RIF and TMP. As our test organisms we will use the reference B. subtilis strain 168 and Staphylococcus epidermidis strain ATCC 12228, a harmless human commensal organism closely related to S. aureus. Central Objectives: 1. Measure mutation rates of B. subtilis and S. epidermidis to RIF and TMP resistance, singly and in combination, in spaceflight and ground control samples. 2. Recover antibiotic-resistant mutants and determine the nucleotide sequence changes in the rpoB and dfrA genes leading to resistance to RIF and TMP respectively. 3. Characterize altered phenotypes of the mutants obtained in Objectives (1) and (2). Methods: Preliminary ground-based studies will be performed using rotating High Aspect Ratio Vessels (HARVs) to simulate microgravity. Spaceflight experiments will be performed using the Biological Research in Canisters-Petri Dish Fixation Units (BRIC-PDFUs) developed for use on the International Space Station with asynchronous ground controls. HARVs, BRIC-PDFUs, and support personnel are conveniently located at the Space Life Sciences Lab, Kennedy Space Center. All microbiology and molecular biology techniques proposed are routinely performed in our laboratory. Significance: The proposed study responds to the NRC’s 2011 Decadal Report Recommendation P2, to: “Establish a robust space flight program of research analyzing plant and microbial growth and physiological responses to the multiple stimuli encountered in space flight environments”, and directly relates to section C2 of the present NRA-“Rapid Turnaround Space Flight Experiments”, specifically addressing the question “What are the genetic changes associated with responses to space flight in biological systems?” Understanding multidrug resistance in opportunistic pathogens during long-term human space flight directly benefits astronaut health and the success of extended missions both in LEO and in interplanetary space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In several ways, astronauts on long-duration space flights share commonalities with patients in long-term clinical care settings: Both groups are relatively immobile, immunocompromised, share close living quarters with others in a similar weakened physiologic condition, and are susceptible to opportunistic infections. Understanding how multiple antibiotic resistance arises in opportunistic human pathogens during space flight can be directly translated to Earth-based medicine.
Task Progress:	During the reporting period, ground-based clinostat studies were undertaken using the organisms Bacillus subtilis and Staphylococcus epidermidis. Both organisms grew to a significantly higher cell density in simulated microgravity than in the 1xg controls. This was not due to increased exponential growth rate, but due to enhanced survival in the stationary phase. B. subtilis demonstrated a significantly increased frequency of mutation to rifampicin resistance in simulated microgravity, but S. epidermidis did not. Both organisms were integrated into Biological Research in Canisters (BRIC) canisters and launched to the International Space Station on the SpaceX 3rd mission to ISS, on April 18, 2014. Samples are currently on orbit and scheduled to return to Earth on May 18, 2014 for post-flight processing.
Bibliography Type:	Description: (Last Updated: 07/25/2025)
Abstracts for Journals and Proceedings	Fajardo-Cavazos P, Narvel R, Nicholson W. "Development of multiple antibiotic resistance in Bacillus subtilis cells exposed to simulated microgravity." Presented at the 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, Nov. 3-8, 2013. 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, Nov. 3-8, 2013. Abstract IP.18. Program and abstracts, p. 122-123. https://asgsr.org/images/stories/pdf/Complete_Program_FINAL_2013_Meeting.pdf , Nov-2013
Abstracts for Journals and Proceedings	Nicholson WL, Narvel R, Moeller R, Fajardo-Cavazos P. "Development of Multiple Antibiotic Resistance in Opportunistic Pathogens Exposed to Spaceflight: the BRIC-18 mission to ISS." Presented at the 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, Nov. 3-8, 2013. 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, Nov. 3-8, 2013. Abstract C4.3. Program and abstracts, p. 52. https://asgsr.org/images/stories/pdf/Complete_Program_FINAL_2013_Meeting.pdf , Nov-2013

Abstracts for Journals and Proceedings	Nicholson WL. "Development of multiple antibiotic resistance by opportunistic bacterial pathogens during human space flight." NASA/ESA/JAXA/CSA ISS Increment 37/38 Science Symposium, May 29, 2013. NASA/ESA/JAXA/CSA ISS Increment 37/38 Science Symposium, May 29, 2013. Remote oral presentation via iLinc, May 2013. , May-2013
Significant Media Coverage	Nicholson WL. "Interviewed by 'NASA Edge' for launch coverage of SpaceX 3rd mission to ISS." NASA Edge Internet video, http://www.ustream.tv/nasaedge , April 2014., Apr-2014